KAIZEN MATHS

Lesson starter

Differential Equations: First and Second Order

First-Order Linear

Name _____

S1

Solve each first-order linear differential equation using an integrating factor.

$\mathrm{d}y\,\mathrm{d}x + 4y = 20$

S2

Solve each first-order linear differential equation and use the initial condition.

dy dx + 3y = 6, with y = 3 when x = 0

S3

Solve each first-order linear differential equation using an integrating factor.

$\mathrm{d}y\,\mathrm{d}x + y = 4x + 2$

S4

Solve each first-order linear differential equation using an integrating factor.

$\mathrm{d}y\,\mathrm{d}x - 2y = e^{4}x$

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Differential Equations: First and Second Order

First-Order Linear

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Solve each first-order linear differential equation using an integrating factor.

$\mathrm{d}y\,\mathrm{d}x + 3y = 15$

E2

Solve each first-order linear differential equation and use the initial condition.

$\mathrm{d}y\,\mathrm{d}x + 4y = 24$, with $y = 9$ when $x = 0$

E3

Solve each first-order linear differential equation using an integrating factor.

$\mathrm{d}y\,\mathrm{d}x + y = 4x + 3$

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Teacher answers & working

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S1

Solve each first-order linear differential equation using an integrating factor.

$dy\,dx + 4y = 20$

Answer:

$y = 5 + Ce^{-4x}$

Working:

This is in the form $y' + P(x)y = Q(x)$, where $P(x) = 4$.

Integrating factor = $e^{\int 4\,dx} = e^{4x}$

Multiply the whole equation by e^{4x}.

$e^{4x}\frac{dy}{dx} + 4e^{4x}y = 20e^{4x}$

The left side is the derivative of ye^{4x}.

$\frac{d}{dx}(ye^{4x}) = 20e^{4x}$

Integrate both sides.

$ye^{4x} = 5e^{4x} + C$

Divide by e^{4x}.

$y = 5 + Ce^{-4x}$

S2

Solve each first-order linear differential equation and use the initial condition.

$dy\,dx + 3y = 6$, with $y = 3$ when $x = 0$

Answer:

$y = 2 + e^{-3x}$

Working:

Integrating factor = $e^{\int 3\,dx} = e^{3x}$

Multiply by e^{3x} and combine the left side.

$\frac{d}{dx}(ye^{3x}) = 6e^{3x}$

Integrate both sides.

$ye^{3x} = 2e^{3x} + C$

Divide by e^{3x}.

$y = 2 + Ce^{-3x}$

Use $y = 3$ when $x = 0$.

$3 = 2 + Ce^{0}$

$C = 1$

$y = 2 + e^{-3x}$

Teacher answers & working

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S3

Solve each first-order linear differential equation using an integrating factor.

$dy\,dx + y = 4x + 2$

Answer:

$y = 4x - 2 + Ce^{-x}$

Working:

Integrating factor = $e^{\int 1\,dx} = e^{x}$

Multiply by e^{x}.

$e^{x}\frac{dy}{dx} + e^{x}y = (4x + 2)e^{x}$

The left side is the derivative of ye^{x}.

$\frac{d}{dx}(ye^{x}) = (4x + 2)e^{x}$

Integrate the right side.

$\int (4x + 2)e^{x}\,dx = e^{x}(4x - 2) + C$

$ye^{x} = e^{x}(4x - 2) + C$

Divide by e^{x}.

$y = 4x - 2 + Ce^{-x}$

S4

Solve each first-order linear differential equation using an integrating factor.

$dy\,dx - 2y = e^{4}x$

Answer:

$y = \frac{1}{2}e^{4x} + Ce^{2x}$

Working:

Write the equation as $y' + P(x)y = Q(x)$, so $P(x) = -2$.

Integrating factor = $e^{\int -2\,dx} = e^{-2x}$

Multiply by e^{-2x}.

$\frac{d}{dx}(ye^{-2x}) = e^{4x}e^{-2x}$

Simplify the right side.

$\frac{d}{dx}(ye^{-2x}) = e^{2x}$

Integrate both sides.

$ye^{-2x} = \frac{1}{2}e^{2x} + C$

Multiply by e^{2x}.

$y = \frac{1}{2}e^{4x} + Ce^{2x}$

Teacher answers & working

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E1

Solve each first-order linear differential equation using an integrating factor.

$dy\,dx + 3y = 15$

Answer:

$y = 5 + Ce^{-3x}$

Working:

This is in the form $y' + P(x)y = Q(x)$, where $P(x) = 3$.

Integrating factor = $e^{\int 3\,dx} = e^{3x}$

Multiply the whole equation by e^{3x}.

$e^{3x}\frac{dy}{dx} + 3e^{3x}y = 15e^{3x}$

The left side is the derivative of ye^{3x}.

$\frac{d}{dx}(ye^{3x}) = 15e^{3x}$

Integrate both sides.

$ye^{3x} = 5e^{3x} + C$

Divide by e^{3x}.

$y = 5 + Ce^{-3x}$

E2

Solve each first-order linear differential equation and use the initial condition.

dy dx + 4y = 24, with y = 9 when x = 0

Answer:

$y = 6 + 3e^{-4x}$

Working:

Integrating factor = $e^{\int 4\,dx} = e^{4x}$

Multiply by e^{4x} and combine the left side.

$\frac{d}{dx}(ye^{4x}) = 24e^{4x}$

Integrate both sides.

$ye^{4x} = 6e^{4x} + C$

Divide by e^{4x}.

$y = 6 + Ce^{-4x}$

Use y = 9 when x = 0.

$9 = 6 + Ce^{0}$

$C = 3$

$y = 6 + 3e^{-4x}$

Teacher answers & working

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E3

Solve each first-order linear differential equation using an integrating factor.

$dy\,dx + y = 4x + 3$

Answer:

$y = 4x - 1 + Ce^{-x}$

Working:

Integrating factor = $e^{\int 1\,dx} = e^x$

Multiply by e^x.

$e^x \frac{dy}{dx} + e^x y = (4x + 3)e^x$

The left side is the derivative of ye^x.

$\frac{d}{dx}(ye^x) = (4x + 3)e^x$

Integrate the right side.

$\int (4x + 3)e^x\,dx = e^x(4x - 1) + C$

$ye^x = e^x(4x - 1) + C$

Divide by e^x.

$y = 4x - 1 + Ce^{-x}$